

Lecture 6. Maximum Principle

Maximum Principle for Discrete Time Optimal Control

- Given a state space $\mathbf{X}$, a control space $\mathbf{U}$, an observation space $\mathbf{Y}$, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$

Value Function and Co-State

- Bellman equation:

$$J_k^*(x_k) = \min_{u_k} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$

- Linear parameterization with co-state λ_k (as the gradient of the value)

$$J_k^*(x_k) = \lambda_k^T x_k + \gamma_k$$

- Relationship between co-states

$$\lambda_k^T x_k + \gamma_k = \min_{u_k} \{l_k(x_k, u_k) + \lambda_{k+1}^T f(x_k, u_k) + \gamma_{k+1}\}$$

Hamiltonian

$$\lambda_k^T x_k + \gamma_k = \min_{u_k} \{ \underbrace{l_k(x_k, u_k) + \lambda_{k+1}^T f(x_k, u_k) + \gamma_{k+1}}_{H_k(x_k, u_k, \lambda_{k+1})} \}$$

Hamiltonian

- Hamiltonian is conceptually equivalent to cost-to-go.

$$J_k(x_k; \underbrace{u_k, u_{k+1} \dots, u_{N-1}}_{\text{future decisions}}) = l_k(x_k, u_k) + \underbrace{\sum_{m=k+1}^{N-1} l_m(x_m, u_m) + l_N(x_N)}_{\text{cost-to-go}}$$

$$\underbrace{H_k(x_k, u_k, \lambda_{k+1})}_{\text{Hamiltonian}} = l_k(x_k, u_k) + \underbrace{\lambda_{k+1}^T f(x_k, u_k)}_{\text{future cost}}$$

- The future decision variables are considered by the co-state.

Dynamic Programming with Co-States

$$\lambda_k^T x_k + \gamma_k = \min_{u_k} \{ \underbrace{l_k(x_k, u_k) + \lambda_{k+1}^T f(x_k, u_k) + \gamma_{k+1}}_{H_k(x_k, u_k, \lambda_{k+1})} \}$$

Hamiltonian

- Assume all functions are smooth
- The optimal solution of the bellman equation above should satisfy

$$\lambda_k = \frac{\partial}{\partial x} H_k(x_k, u_k, \lambda_{k+1}) = \frac{\partial}{\partial x} l_k(x_k, u_k) + \left[\frac{\partial}{\partial x} f(x_k, u_k) \right]^T \lambda_{k+1}$$

$$0 = \frac{\partial}{\partial u} H_k(x_k, u_k, \lambda_{k+1}) = \frac{\partial}{\partial u} l_k(x_k, u_k) + \left[\frac{\partial}{\partial u} f(x_k, u_k) \right]^T \lambda_{k+1}$$

$$u_k = \arg \min_u H_k(x_k, u, \lambda_{k+1})$$

Dynamic Programming with Co-States

- At time N , we find a co-state function $\lambda_N(x_N)$ and a γ_N that satisfies

$$\lambda_N(x_N) \cdot x_N + \gamma_N(x_N) = l_N(x_N), \quad \lambda_N(x_N) = \nabla_x l_N(x_N)$$

- For $k = N - 1, N - 2, \dots, 1, 0$, we solve the following equations

$$\lambda_k = \frac{\partial}{\partial x} H_k(x_k, u_k, \lambda_{k+1}) = \frac{\partial}{\partial x} l_k(x_k, u_k) + \left[\frac{\partial}{\partial x} f(x_k, u_k) \right]^T \lambda_{k+1}$$

maybe a function of $x_{k+1} = f(x_k, u_k)$

$$u_k = \arg \min_u H_k(x_k, u, \lambda_{k+1})$$

to obtain the optimal control policy $u_k = q_k^*(x_k)$ and co-state function $\lambda_k(x_k)$. The constant term should still be computed from the Bellman equation.

Dynamic Programming with Co-States

- The optimal control sequence and co-states can be obtained in a forward pass considering the initial state x_0

$$x_0^* = x_0$$

$$u_k^* = q_k^*(x_k^*), \forall k = 0, 1, \dots, N-1$$

$$\lambda_k^* = \lambda_k(x_k^*), \forall k = 0, 1, \dots, N$$

$$x_{k+1}^* = f(x_k^*, u_k^*), \forall k = 0, 1, \dots, N-1$$

How to Interpret the Co-State?

- We introduced co-state as one way to parameterize the value function and define it as the gradient of the value function.
- It can also be interpreted as:
 - Shadow price
 - Lagrange multiplier

Co-State and Shadow Price

$$J_k^*(x_k) = \lambda_k^T x_k + \gamma_k$$

- Shadow price
 - The “unit cost” of the current state value
 - Will be illustrated later

Co-State and Lagrange Multiplier*

- The optimization problem (primal):

$$\min_{u_k, \boxed{x_{k+1} = f(x_k, u_k)}} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$

- Lagrangian function with multiplier

$$L(x_k, u_k, x_{k+1}, \lambda_{k+1}) = l_k(x_k, u_k) + J_{k+1}^*(x_{k+1}) - \lambda_{k+1}^T (x_{k+1} - f(x_k, u_k))$$

- Lagrangian dual function

$$L^*(x_k, \lambda_{k+1}) = \min_{u_k, x_{k+1}} L(x_k, u_k, x_{k+1}, \lambda_{k+1})$$

Co-State and Lagrange Multiplier*

- The maximization of the dual problem provides a lower bound to the solution of the primal problem

$$\max_{\lambda_{k+1}} \min_{u_k, x_{k+1}} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1}) - \lambda_{k+1}^T (x_{k+1} - f(x_k, u_k))\} \leq \min_{u_k, x_{k+1} = f(x_k, u_k)} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$

$$L^*(x_k, \lambda_{k+1})$$

- The difference is called the duality gap.
- There is no equality constraint in the dual form. Hence it is easier to find a solution for the dual problem.

Co-State and Lagrange Multiplier*

$$\max_{\lambda_{k+1}} \min_{u_k, x_{k+1}} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1}) - \lambda_{k+1}^T (x_{k+1} - f(x_k, u_k))\} \leq \min_{u_k, x_{k+1} = f(x_k, u_k)} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$



$$\max_{\lambda_{k+1}} \left\{ \min_{u_k} \{ \underline{l_k(x_k, u_k) + \lambda_{k+1}^T f(x_k, u_k)} \} + \min_{x_{k+1}} \{ \underline{J_{k+1}^*(x_{k+1}) - \lambda_{k+1}^T x_{k+1}} \} \right\}$$

Hamiltonian

According to the first-order optimality condition, the optimal solution $x_{k+1}^*, \lambda_{k+1}^*$ should satisfy

$$\frac{\partial}{\partial x} J_{k+1}^*(x_{k+1}^*) = \lambda_{k+1}^*$$

This justifies the linear parameterization

Maximum Principle

- Theorem:
 - If $\{u_0^*, \dots, u_{N-1}^*\}$ and $\{x_1^*, \dots, x_N^*\}$ are the optimal state and control trajectories starting at x_0 , then there exists a co-state trajectory $\{\lambda_1^*, \dots, \lambda_N^*\}$ with $\lambda_N^* = \frac{\partial}{\partial x_N} J_N^*(x_N^*)$ satisfying

$$x_{k+1}^* = \frac{\partial}{\partial \lambda} H_k(x_k^*, u_k^*, \lambda_{k+1}^*) = f(x_k^*, u_k^*)$$

$$\lambda_k^* = \frac{\partial}{\partial x} H_k(x_k^*, u_k^*, \lambda_{k+1}^*) = \frac{\partial}{\partial x} l_k(x_k^*, u_k^*) + \left[\frac{\partial}{\partial x} f(x_k^*, u_k^*) \right]^T \lambda_{k+1}^*$$

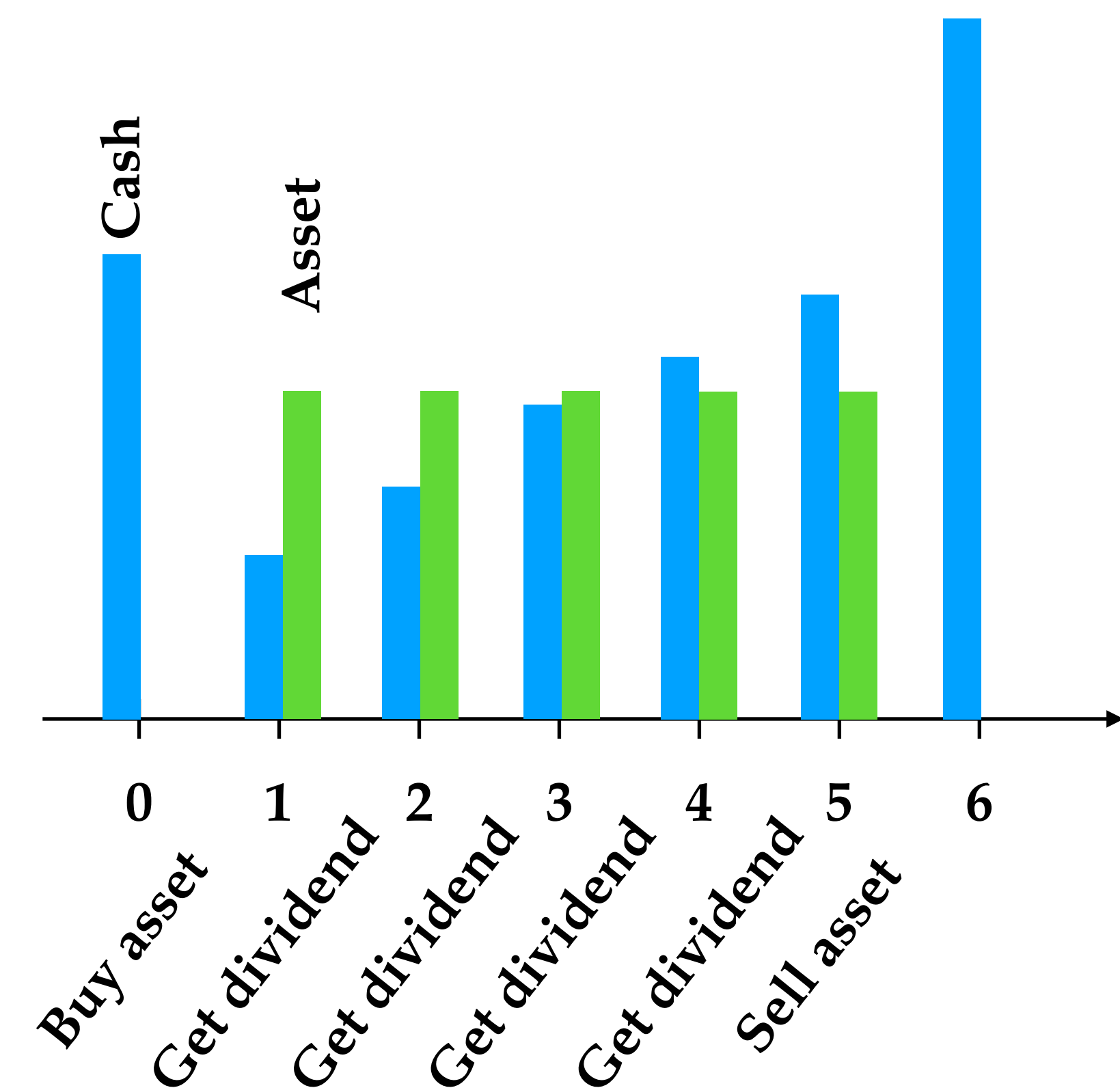
$$u_k^* = \arg \min_u H_k(x_k^*, u, \lambda_{k+1}^*)$$

Maximum Principle

- It is a necessary condition, not a sufficient condition.
- The solutions we obtained using “dynamic programming with co-states” always satisfy the maximum principle.

Example: Investment Problem

- I have \$m cash and want to maximize my cash 6 month later.
- There is an asset A on the market with monthly dividend. A's price is floating and external to me (I cannot influence the price).
- I can choose to trade in/out asset A at the beginning of every month.
- If I know the future prices of A, what is my trading strategy to maximize my cash 6 month later?

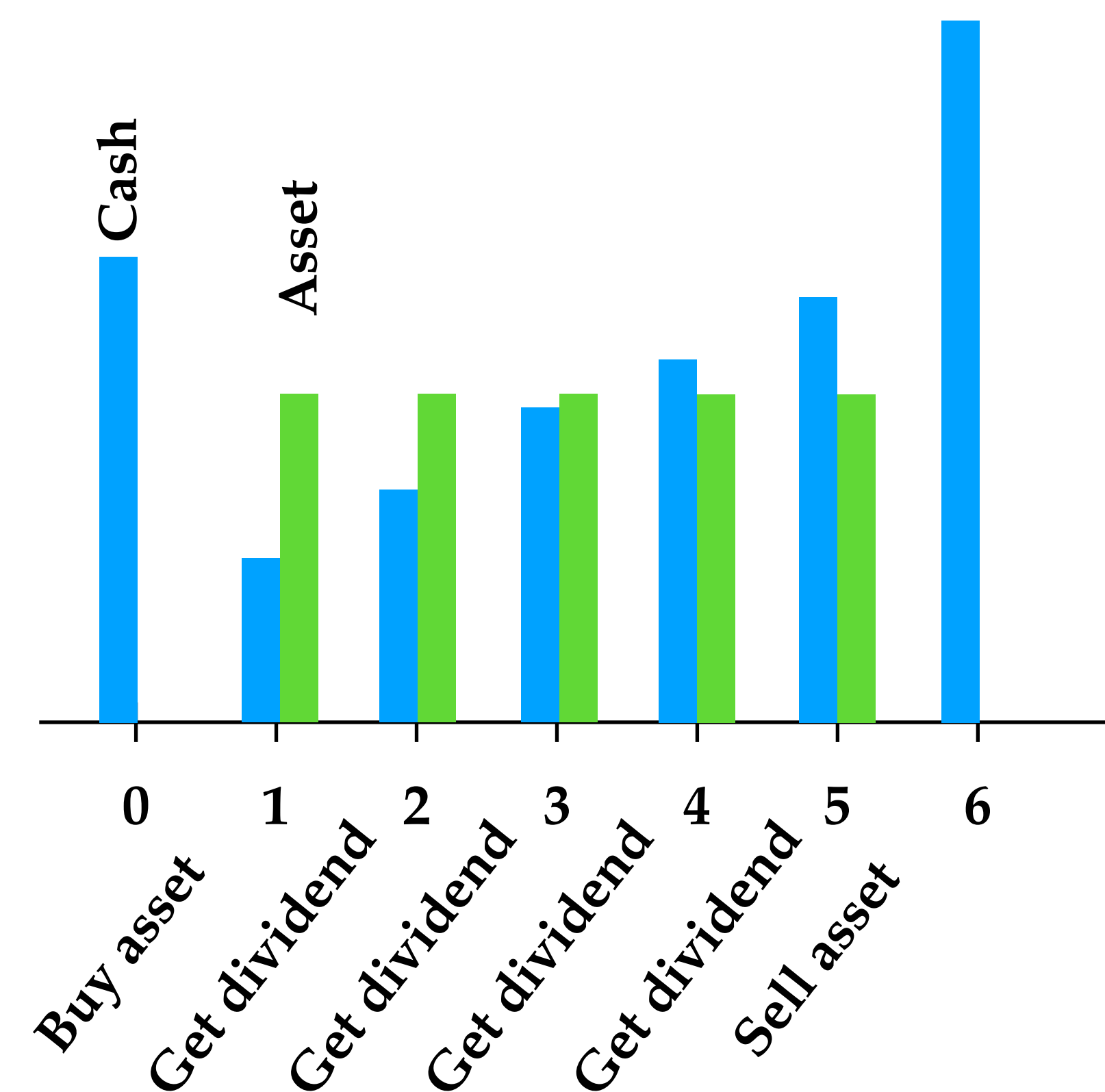


Example: Investment Problem

- State: the amount of cash and the amount of asset $x \in \mathbb{R}^2$, with $x_0 = [m \ 0]^T$
- Control: the amount of asset bought $u \in [-1, 1]$
- Dynamics:

$$x_{k+1} = \begin{bmatrix} 1 & r_k \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} -p_k \\ 1 \end{bmatrix} u_k$$

Dividend rate at month k
Price at month k



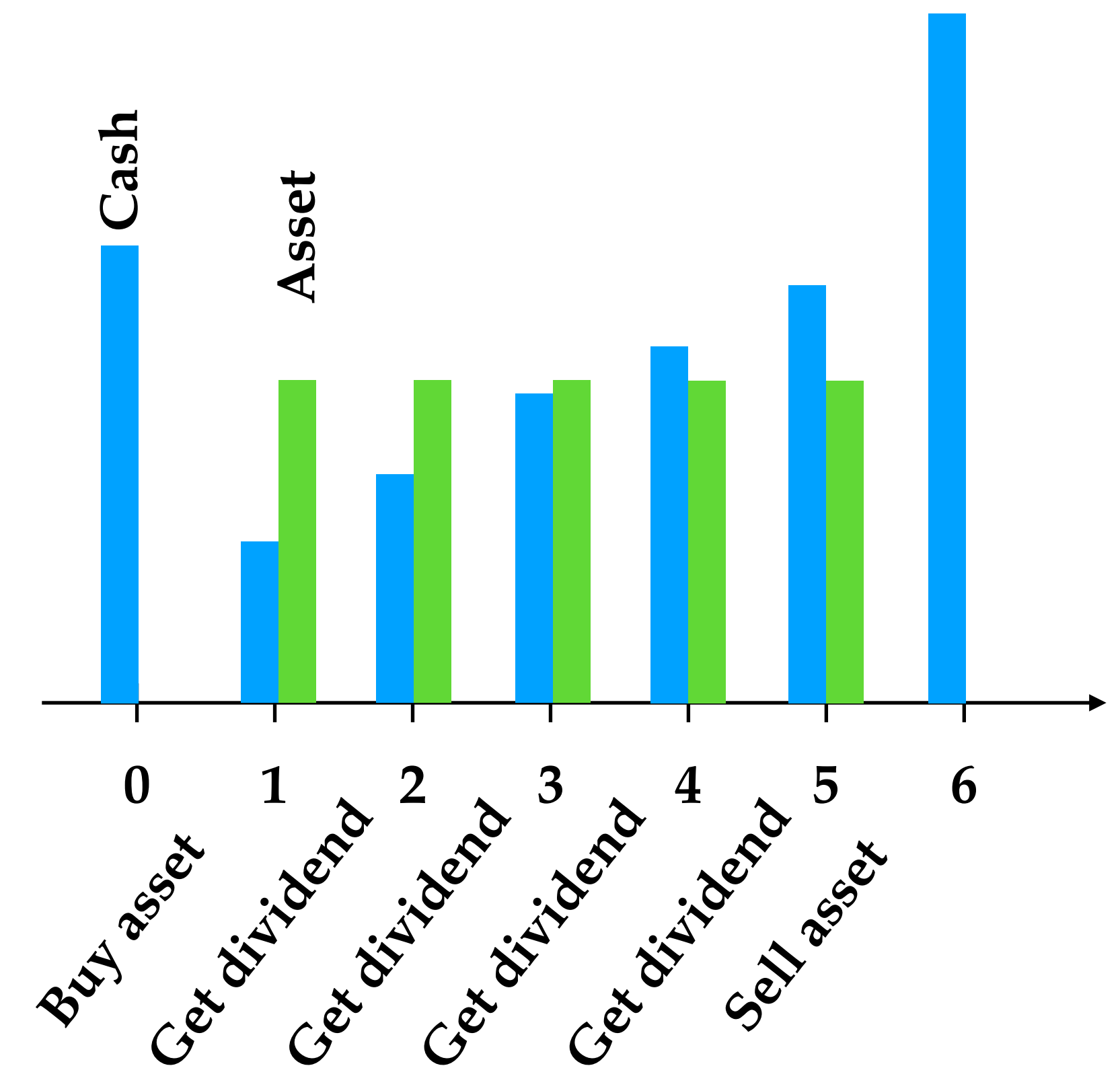
Example: Investment Problem

- No run-time cost
- We are minimizing

$$J = \begin{bmatrix} -1 & 0 \end{bmatrix} x_6$$

- Control constraints to ensure the problem is bounded

$$u \in [-1, 1]$$



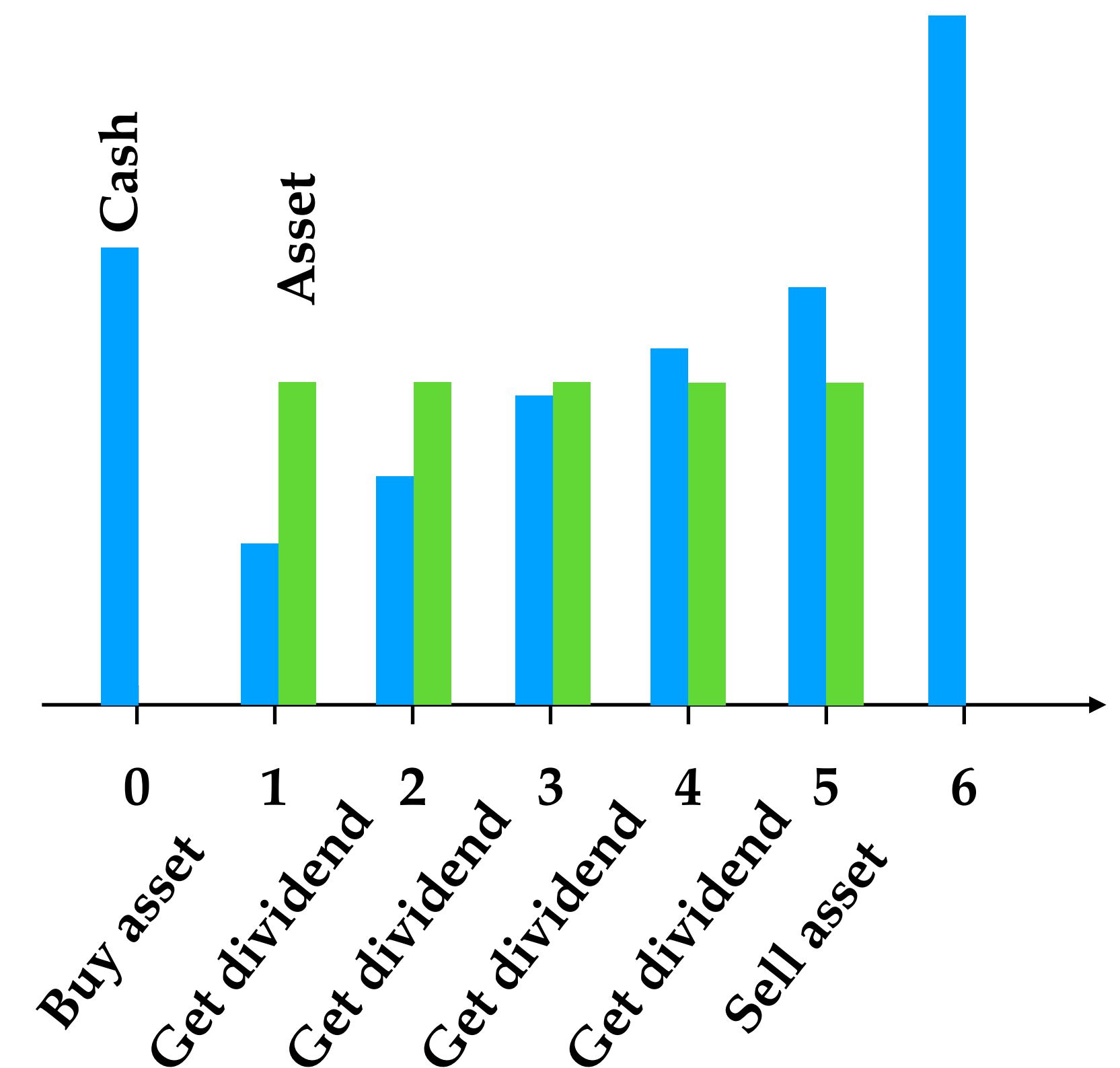
Example: Investment Problem

- State: the amount of cash and the amount of asset $x \in \mathbb{R}^2$, with $x_0 = [m \ 0]^T$
- Control: the amount of asset bought $u \in [-1, 1]$
- Dynamics:

$$x_{k+1} = \begin{bmatrix} 1 & r_k \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} -p_k \\ 1 \end{bmatrix} u_k$$

- Objective

$$J = \begin{bmatrix} -1 & 0 \end{bmatrix} x_6$$



Example: Investment Problem

- Value function: $J_k^*(x_k) = \lambda_k^T x_k + \gamma_k$
- Since there is no run-time cost, $J_k^*(x_k) = J_6^*(x_6)$
- Shadow price: How much does cash/asset **worth** at month k with respect to the objective function?
- For example, $\lambda_6 = [-1 \ 0]^T$ means that asset worths nothing at month 6.
- We need to solve for the shadow price at every stage.

Example: Investment Problem

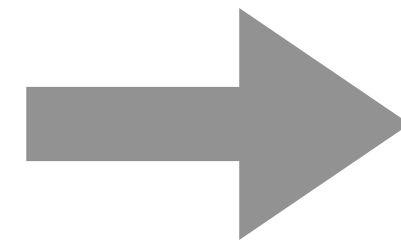
- Hamiltonian

$$H_k = \lambda_{k+1}^T \left\{ \begin{bmatrix} 1 & r_k \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} -p_k \\ 1 \end{bmatrix} u_k \right\}$$

State =
[amount of cash;
amount of asset]

Co-state =
[shadow price of cash;
shadow price of asset]

$$\lambda_k = \frac{\partial}{\partial x} H_k(x_k, u_k, \lambda_{k+1})$$



$$\lambda_k = \begin{bmatrix} \boxed{1} & \boxed{0} \\ r_k & 1 \end{bmatrix} \lambda_{k+1}$$

The shadow price of cash does not change throughout time

The shadow price of asset is higher going backward in time, due to dividend payoff in the future.

Example: Investment Problem

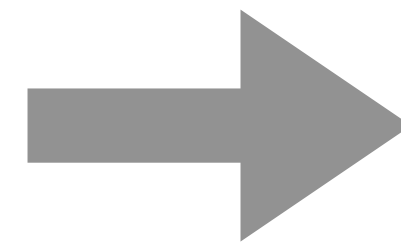
- Hamiltonian

$$H_k = \lambda_{k+1}^T \left\{ \begin{bmatrix} 1 & r_k \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} -p_k \\ 1 \end{bmatrix} u_k \right\}$$

State =
[amount of cash;
amount of asset]

Co-state =
[shadow price of cash;
shadow price of asset]

$$u_k = \arg \min_u H_k(x_k, u, \lambda_{k+1})$$



$$u_k = \begin{cases} -1 & [-p_k \ 1] \lambda_{k+1} \geq 0 \\ 1 & [-p_k \ 1] \lambda_{k+1} < 0 \end{cases}$$

The payoff in the future if we buy one unit of asset:
we will get one more unit of asset, hence gain $\lambda_{k+1,2}$ cost
we will use p_k amount of cash, hence reduce $p_k \lambda_{k+1,1}$ cost

Control strategy: buy asset if it costs less, otherwise sell

Example: Investment Problem

- Control strategy:

$$u_k = \begin{cases} -1 & [-p_k \ 1] \lambda_{k+1} \geq 0 \\ 1 & [-p_k \ 1] \lambda_{k+1} < 0 \end{cases}$$

- Buy asset if it costs less, otherwise sell
- This is a bang-bang control
- The switch depends on the future perspective of the payoff, i.e., the shadow price

Example: Investment Problem

$$\lambda_k = \begin{bmatrix} 1 & 0 \\ r_k & 1 \end{bmatrix} \lambda_{k+1} \quad u_k = \begin{cases} -1 & [-p_k \ 1] \lambda_{k+1} \geq 0 \\ 1 & [-p_k \ 1] \lambda_{k+1} < 0 \end{cases}$$

Month	Dividend rate	Asset price	Co-state (shadow price)	Action
6			[-1, 0]	
5	1	1	[-1, -1]	-1
4	0	2	[-1, -1]	-1
3	1	3	[-1, -2]	-1
2	0	3	[-1, -2]	-1
1	1	2	[-1, -3]	1
0	0	1	[-1, -3]	1

Example: Investment Problem

- This is a constrained linear time-varying problem.
- Solutions to linear optimal control problems are usually bang-bang control.
- In linear optimal control problems, the co-state usually does not depend on the state. But in linear quadratic problems, the co-state depends on the state.
- In this example, it is assumed that we know the future dynamics, i.e., future price and dividend rate. In the case that the dynamics are unknown, the problem becomes stochastic, which requires adaptive strategies.

Maximum Principle for Continuous Time Optimal Control

- Given a state space $\mathbf{X}$, a control space $\mathbf{U}$, an observation space $\mathbf{Y}$, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$

Connection between HJB and Hamiltonian

- The Hamiltonian in the continuous time is defined as

$$H(t, x_t, u_t, \lambda_t) := l_t(x_t, u_t) + \lambda_t^T f(x_t, u_t)$$

- The co-state is the gradient of the value function

$$\lambda_t = \nabla_x V(t, x_t)$$

- The Hamiltonian is exactly the term inside the minimization in HJB

$$\dot{V}(t, x_t) = - \min_{u_t} \{ l_t(x_t, u_t) + \nabla_x V(t, x_t) \cdot f(x_t, u_t) \}$$

Connection between HJB and Hamiltonian

- Similar to the derivation in discrete time case, the co-state λ_t is the Lagrangian multiplier for the dynamic equation $\dot{x} = f(x, u)$
- How does the co-state evolve in time?

Dynamics of the Co-State

$$\dot{V}(t, x_t) = - \min_{u_t} \{ l_t(x_t, u_t) + \nabla_x V(t, x_t) \cdot f(x_t, u_t) \}$$

Using the Partial Derivative Lemma, under optimal control u_t

$$-\frac{\partial^2}{\partial x \partial t} V(t, x_t) = \frac{\partial}{\partial x} l_t(x_t, u_t) + \frac{\partial^2}{\partial x^2} V(t, x_t) f(x_t, u_t) + \left[\frac{\partial}{\partial x} f(x_t, u_t) \right]^T \frac{\partial}{\partial x} V(t, x_t)$$

Using the following identity

$$\frac{d}{dt} \frac{\partial}{\partial x} V(t, x_t) = \frac{\partial^2}{\partial t \partial x} V(t, x_t) + \frac{\partial^2}{\partial x^2} V(t, x_t) f(x_t, u_t)$$

We have

$$-\frac{d}{dt} \frac{\partial}{\partial x} V(t, x_t) = \frac{\partial}{\partial x} l_t(x_t, u_t) + \left[\frac{\partial}{\partial x} f(x_t, u_t) \right]^T \frac{\partial}{\partial x} V(t, x_t)$$

λ_t λ_t

Pontryagin's Maximum Principle

- Theorem:
 - If $x_{0:T}^*$ and $u_{0:T}^*$ are the optimal state and control trajectories starting at x_0 , then there exists a co-state trajectory $\lambda_{0:T}^*$ with $\lambda_T^* = \frac{\partial}{\partial x} l_T(x_T^*)$ satisfying

$$\begin{aligned}\dot{x}_t^* &= \frac{\partial}{\partial \lambda} H(t, x_t^*, u_t^*, \lambda_t^*) = f(x_t^*, u_t^*) \\ -\dot{\lambda}_t^* &= \frac{\partial}{\partial x} H(t, x_t^*, u_t^*, \lambda_t^*) = \frac{\partial}{\partial x} l_t(x_t^*, u_t^*) + \left[\frac{\partial}{\partial x} f(x_t^*, u_t^*) \right]^T \lambda_t^* \\ u_t^* &= \arg \min_u H(t, x_t^*, u, \lambda_t^*)\end{aligned}$$

Maximum Principle in Continuous Time vs. Discrete Time

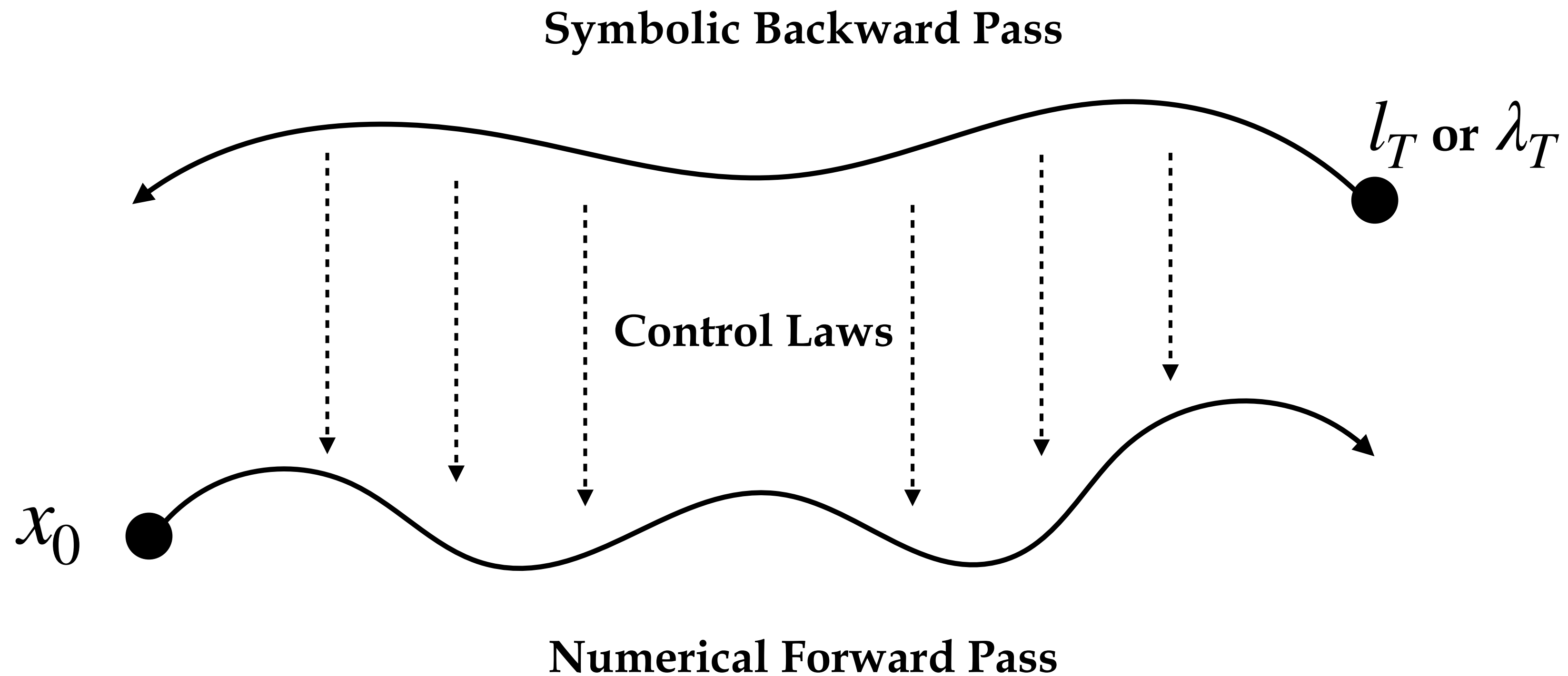
Continuous time

$$\begin{aligned}
 H(t, x_t, u_t, \lambda_t) &:= l_t(x_t, u_t) + \lambda_t^T f(x_t, u_t) \\
 \dot{x}_t^* &= \frac{\partial}{\partial \lambda} H(t, x_t^*, u_t^*, \lambda_t^*) = f(x_t^*, u_t^*) \\
 -\dot{\lambda}_t^* &= \frac{\partial}{\partial x} H(t, x_t^*, u_t^*, \lambda_t^*) = \frac{\partial}{\partial x} l_t(x_t^*, u_t^*) + \left[\frac{\partial}{\partial x} f(x_t^*, u_t^*) \right]^T \lambda_t^* \\
 u_t^* &= \arg \min_u H(t, x_t^*, u, \lambda_t^*)
 \end{aligned}$$

Discrete time

$$\begin{aligned}
 H_k(x_k, u_k, \lambda_{k+1}) &= l_k(x_k, u_k) + \lambda_{k+1}^T f(x_k, u_k) \\
 x_{k+1}^* &= \frac{\partial}{\partial \lambda} H_k(x_k^*, u_k^*, \lambda_{k+1}^*) = f(x_k^*, u_k^*) \\
 \lambda_k^* &= \frac{\partial}{\partial x} H_k(x_k^*, u_k^*, \lambda_{k+1}^*) = \frac{\partial}{\partial x} l_k(x_k^*, u_k^*) + \left[\frac{\partial}{\partial x} f(x_k^*, u_k^*) \right]^T \lambda_{k+1}^* \\
 u_k^* &= \arg \min_u H_k(x_k^*, u, \lambda_{k+1}^*)
 \end{aligned}$$

Two-Point Boundary Value Problem



Exercise

- Dynamics:

$$\dot{x} = u, |u| \leq 1, t \in [0, T]$$

- Cost function

$$J(x_0) = \min_u |x(T)|$$

- What is the Hamiltonian? What is the co-state trajectory?